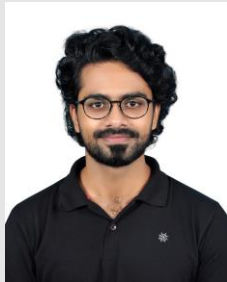


Dr. AKHIL K. RAMESH



EMAIL: -

akhil.k.ramesh@gmail.com
ark@ieee.org

PHONE

+91- 9495637750
+65- 89306204

PERMANENT ADDRESS:

Kurumathu (House),
Karimannoor (P.O),
Idukki District,
Kerala – 685581,
India.

DATE OF BIRTH

27TH March 1990

NATIONALITY

Nationality : Indian

TRAVEL HISTORY

Taiwan (*Gold Card Holder*),
Singapore(*E-Pass*), South
Korea, Australia, Thailand,
Macau, Nepal, Philippines,
Malaysia, USA

PROFESSIONAL BODIES

1. IEEE Member (94903253)
2. IEEE Electron Devices Society Member
3. IEEE Magnetics Society Member
4. IEEE Nanotechnology Council
5. IEEE Sensors Council
6. Micro Electro Mechanical Systems Technical Community.
7. IEEE Education Society

LANGUAGES

Malayalam : Native
English : Excellent
Hindi : Excellent
Chinese : Beginner

RESEARCH AREA

- Spintronics
- Flexible Electronics
- Nanofabrication
- MEMS

EDUCATION

2017 – 2022	PhD (Development of Spintronic Devices for Sensing & Memory Applications) Indian Institute of Technology Delhi, India National Yang Ming Chiao Tung University, Hsinchu, Taiwan. (IITD-NYCU Joint Degree Program) Grade point = 8 / 10
2013 – 2015	M.Tech in Electronics (VLSI & embedded Systems) Cochin University of Science and Technology, Kerala, India. Grade point = 9.24 / 10
2008 – 2012	B.Tech in Electronics and Instrumentation Engineering. Cochin University of Science and Technology, Kerala, India. Percentage = 69.64%
2008	Higher Secondary Certificate (12 th) (CBSE), Jawahar Navodaya Vidyalaya Idukki, Kerala, India. Percentage = 76.20%
2006	High School Certificate (10 th) (CBSE), Jawahar Navodaya Vidyalaya Idukki, Kerala, India. Percentage = 85.40%

PROFESSIONAL EXPERIENCE

May 2024 – Present	Research Fellow NTI-NTU Corp Lab, Nanyang Technological University, Singapore
Aug 2022 - May 2024	Research Scientist Temasek Lab, Nanyang Technological University, Singapore
Sep 2020 - Jan 2021	Graduate Teaching Assistant International College of Semiconductor Technology, NYCU, Taiwan.
Jul 2017 – Feb 2022	Graduate Research Student IIT Delhi and NYCU Taiwan
Jan 2017 - Jul 2017	Project associate Center for applied research in Electronics (CARE), IIT Delhi, India
Jul 2015 - Jan 2017	Assistant Professor Department of ECE, College of Engineering Munnar, Kerala, India
Jul 2012 - Apr 2013	Lecturer Department of ECE, College of Engineering Munnar, Kerala, India

RESEARCH EXPERTISE

- PhD Thesis : **Interface Engineering in CoFeB based Spintronic Devices for Sensing and Memory Applications**
- Advisors : Prof. Pushparaj Singh (IIT Delhi) & Prof. Yuan-Chieh Tseng (NYCU,Taiwan)
1. **Fabrication and development of Spintronic MEMS sensors**
 - ✓ Design, fabrication, characterization and testing of spintronic sensors for Microelectromechanical systems and bio sensing applications
 - ✓ Fabrication and testing of magnetostrictive strain sensor on flexible substrate
 2. **Interface Engineering of Magnetic Tunnel Junctions used for STT-MRAM applications**
 - ✓ Studying the effect of interface roughness in CoFeB/MgO heterostructure using atomistic and micromagnetic simulations
 - ✓ Optimizing the CoFeB/MgO heterostructure for achieving Perpendicular magnetic anisotropy (PMA)
 - ✓ Testing and analysis of Mg-inserted Magnetic Tunnel Junctions (MTJ) for performance trade-off study by magneto transport measurements.

TECHNICAL EXPERTISE

1. EXPERIMENTAL

- ✓ Certified user at Taiwan Semiconductor Research Institute (TSRI)
- ✓ Working experience in **Industry standard clean rooms (class 100 and class 1000)**
- ✓ Deposition
 - **RF/ DC Sputtering** Alcatel, Edward, Excel, APT)
 - **Diffusion/Oxidation/CVD furnace** (Tempress and Thermoco)
 - **E-beam/Thermal evaporation System** (HHV, Japan ULVAC EBX-6D and One indigenously build at CARE, IIT Delhi)
- ✓ Lithography
 - **Two Photon Polymerization** (Microlight 3D)
 - **Laser Interface Lithography**
 - **Mask aligner** (ABM,Keyi Technology AG1000-6N-ST and AG1000-4D-DSMV)
 - **Maskless Lithography** systems like **Laser writer** (Microtech LW-405) , **Electron Beam lithography** (Elionix ELS-7500EX)
- ✓ Etching
 - **RIE and ICPRIE** (NEXTRAL NE110, Plasma therm and Oxford)
 - **Ion beam etching (IBE)**
- ✓ Characterization & Measurement
 - **TEM** (JEOL 2200FS)
 - **XRD** (Bruker, Shimadzu)
 - **XPS** (Shimadzu Kratos Axis Supra)
 - **FESEM and FIB** (ZEISS)
 - **AFM** (Park NX10)
 - **VSM/SQUID** (Quantum Design)
 - **Magneto transport measurement**
 - Kiethley parameter analyzer and lakeshore probe station
- ✓ Packaging and Assembly
 - **Wire bonding**
 - Assembly of micromechanical parts of MEMS for testing

2. SIMULATION

- ✓ Magnetic simulation and transport model
 - **OOMMF, Vampire atomistic modelling,**
 - **Wolfram Mathematica**
- ✓ Device modelling and implementation
 - **COMSOL Multiphysics, TCAD.**
 - **Cadence, Xilinx, P-SPICE.**
- ✓ Layout design
 - **Clewin layout editor**

PUBLICATIONS

1. **Ramesh AK**, XingYu Chen, Seetoh IP, Tan WX, Lai C. Polydopamine Assisted Electroless Deposition of Strongly Adhesive NiFe Films for Flexible Spintronics. *ACS Applied Materials & Interfaces*. 2025; 17(8),
2. Seetoh IP, **Ramesh AK**, Tan WX, Lai C. Enhanced Photoelectrochemical Water Splitting using Carbon Cloth Functionalized with ZnO Nanostructures via Polydopamine Assisted Electroless Deposition. *Nanoscale*. 2024;16(17):8401-16.
3. Anbalagan AK, Hu FC, Chan WK, Gandhi AC, Gupta S, Chaudhary M, Chuang KW, **Ramesh AK**, Billo T, Sabbah A, Chiang CY. Gamma-Ray Irradiation Induced Ultrahigh Room-Temperature Ferromagnetism in MoS2 Sputtered Few-Layered Thin Films. *ACS nano*. 2023 Mar 23;17(7):6555-64.
4. Rana V, Gangwar P, **Ramesh AK**, Sharma T, Bhat KN, Nayak MM, Das S, Singh P. High-Performing Polycrystalline MoS2-Based Microelectromechanical Piezoresistive Pressure Sensor. *IEEE Sensors Journal*. 2022 Oct;22(19):18542-9.
5. **Ramesh, A.K.**, Chou, Y.T., Lu, M.T., Singh, P. and Tseng, Y.C., 2022. Biological Sensing Using Anomalous Hall Effect Devices. *Nanotechnology*. 2022 May 3;33(33): 335502.
6. **Ramesh AK**, Cheng CW, Ku TC, Rana V, Gangwar P, Singh P, Tseng YC. Interface Imperfection Effects on Spin Transfer Torque Switching: an Atomistic Approach. *Journal of Physics D: Applied Physics*. 2022 Feb 28;55(21):215002.

7. Chung JC, Kumar Anbalagan A, Fan CL, Liao YH, **Ramesh AK**, Gupta S, Tseng YC, Tai NH, Lee CH. Investigating the mechanism of magnetic phase transition temperature of FeRh thin films by doping copper impurities. *Materials Chemistry and Physics*. 2022 Jan 1; 275:125252.
8. **Ramesh AK**, Chen KM, Lin YJ, Singh P, Wei JH, Hsin YC, Wu CI, Tseng YC. Insertion Trade-off Effects on the Spin-Transfer Torque Memory Explored by In Situ X-ray. *ACS Applied Electronic Materials*. 2021 Aug 23; 3(9):4047-55.
9. Rana V, Gangwar P, Meena JS, **Ramesh AK**, Bhat KN, Das S, Singh P. A highly sensitive wearable flexible strain sensor based on polycrystalline MoS₂ thin film. *Nanotechnology*. 2020 Jul 3; 31(38):385501.
10. Rana V, Ahmad G, **Ramesh AK**, Das S, Singh P. Diameter-dependent piezoresistive sensing performance of junction less gate-all-around nanowire FET. *IEEE Transactions on Electron Devices*. 2020 May 11; 67(7):2884-9.
11. Rana V, Gangwar P, **Ramesh AK**, Chauhan SS, Shrivastava PK, Das S, Singh P. Poly-Crystalline MoS₂ Thin Film Based Low Cost Flexible Sensors. In 2020 *5th IEEE International Conference on Emerging Electronics (ICEE)* 2020 Nov 26 (pp. 1-4). IEEE.
12. **Ramesh AK**, Rana V, Das P, Singh P. Polycrystalline Sense Layer for Magnetic Tunnel Junction (MTJ) as Ultrasensitive Sensing Element for MEMS Pressure Sensors. In 2019 *IEEE 32nd International Conference on Micro Electro Mechanical Systems (MEMS) 2019* Jan 27 (pp. 99-102). IEEE.
13. Rana V, Singh A, **Ramesh AK**, Dhyani V, Das S, Singh P. Sidewall Transfer patterning-based nano-crystalline MoS₂ sensing element for stress and optical MEMS sensor. In 2019 *IEEE 32nd International Conference on Micro Electro Mechanical Systems (MEMS) 2019* Jan 27 (pp. 636-639). IEEE.
14. **Ramesh AK**, Singh P, Das P. Effect of Annealing on Magnetostrictive Sense Layer for Ultrasensitive Spintronic MEMS Sensor. In 2018 *15th IEEE India Council International Conference (INDICON) 2018* Dec 16 (pp. 1-4). IEEE.
15. Krishnan JR, **Ramesh AK**, Indhu KK, Ramesh P. A novel equivalent circuit for analysing the effect of local hotspots in micro PEM fuel cells. *Procedia Technology*. 2016 Jan 1; 25:635-42.
16. **Ramesh AK**, Ramesh P. Trade-off between sensitivity and dynamic range in designing MEMS capacitive pressure sensor. *TENCON 2015-2015 IEEE Region 10 Conference* 2015 Nov 1 (pp. 1-3). IEEE.

PROJECTS & FUNDINGS

1. TL Innovation Grant 2023: Principal investigator for the project based on Spin Seebeck Effect
[Apr. 2023 – Oct. 2023]

TALKS & PRESENTATIONS

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| 1. Invited Talk on “Spin Degree of Freedom Unlocks Nano to Quantum” at Material Science and Engineering Graduate Student Club, NTU Singapore. | Sep. 2024 |
| 2. MMM 2023 , 68th Annual Conference On Magnetism and Magnetic Materials” Electroless Deposition of NiFe on Flexible Substrate: Towards Wearable Magnetic devices for High Temperature Applications.” | Nov. 2023 |
| 3. Case study “Academic Collaborations Leading to Joint/Dual Degrees” at the Global Symposium on Deepening International Collaborations supported by <i>IEEE Education Society</i> | Jun. 2023 |
| 4. Invited Talk on “Advancements in Technology for Interdisciplinary Research” at Kerala APJ Abdul Kalam Technological University sponsored Faculty Development Program at Universal Engineering College, Thrissur, Kerala. | Aug. 2022 |
| 5. Invited Talk on “Opportunities of B.Tech courses beyond just IT jobs” at Yuvavani All India Radio Kochi FM | Aug. 2022 |
| 6. Invited Talk on “Converting Projects into Publications ” for the webinar held under the auspices of Department of Electronics and Communication Engineering, Universal Engineering College, Thrissur, Kerala. | Jul. 2021 |
| 7. Invited Talk on “Spintronics” at Christ College of Engineering, Irinjalakuda, Kerala | |
| 8. INTERMAG 2021 , IEEE Magnetics Society Conference,” Virtualizing CoFeB/MgO Reconstruction Effects on the STT-MTJ’s Performance” | Apr. 2021 |

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| 9. MEMS'19 , IEEE International Conference on MEMS at Seoul, South Korea | Jan. 2019 |
| 10. INDICON 2018 , Coimbatore, India. "Effect of Annealing on Magnetostrictive Sense Layer for Ultrasensitive Spintronic MEMS Sensor" | Dec. 2018 |
| 11. IWPSD , New Delhi, India." Magnetostrictive Sense Layer for MTJ based Ultrasensitive MEMS Pressure Sensor". | Dec. 2017 |
| 12. COMSOL Conference Bangalore 2016 ." FEM analysis of MEMS capacitive pressure sensor with segmented boss structure for diaphragm". | Oct. 2016 |
| 13. TENCON 2015 , IEEE region 10 conference at Macau, China | Nov. 2015 |
| 14. INSTRUMENTANIA 2015 , National Conference on Instrumentation and Control Engineering ""Design of Segmented Boss Structure for MEMS Capacitive Sensor Diaphragm", | Aug. 2015 |
| 15. National Conference on Recent Advancements in Electronic Communication and Allied Areas 2015 , "Range analysis of MEMS Capacitive Pressure Sensor with Bossed Diaphragm" | Feb. 2015 |

AWARDS & ACHIEVEMENTS

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| 1. MediaTek Foundation Scholarship winner at NCTU Taiwan. | Nov. 2019 |
| 2. Scholarship for Outstanding New Student Award from International College of Semiconductor Technology, NYCU, Taiwan (2019-2021). | Sep. 2019 |
| 3. Research Scholar Travel Award form IIT Delhi (2019) | Jan. 2019 |
| 4. Listed among the top 10 exhibits visited by President of India H.E Pratibha Patil, in 34th Jawaharlal Nehru National Science Exhibition-2007, | Dec. 2007 |
| 5. 2 nd prize winner at National level Science exhibition 2006. | Oct. 2006 |

REVIEWER EXPERIENCE

1. Reviewer for Journal of Micromechanics and Microengineering (JMM)
2. Reviewer for Microelectronic engineering
3. Reviewer for Journal of physics communications
4. Reviewer for Discover Nano
5. Reviewer for SPIN

LEADERSHIP AND MANAGEMENT ROLES

1. Represented nanofabrication team for NTI-NTU Corporate lab launch.
2. Asset in charge and Goods Receiving Officer for AMDS group in Temasek Lab @ NTU.
3. Mentored Student Projects
 - ✓ Atomistic simulation of FM/HM heterostructures [NTU Bachelors Final Year Project (FYP)]
 - ✓ Developing an Anomalous Hall Effect Biosensor for Ovarian Biomarker with ELISA [NYCU master's thesis]
 - ✓ Investigation of the atomic micro-magnetic model of the CoFeB/MgO thin film [NYCU master's thesis]
 - ✓ A Novel system for enhancing the lifetime of a sustainable Micro Fuel Cell based power source [CEM Masters]
 - ✓ Analyzing the Effect of Local Hotspots in Micro PEM Fuel Cells [CEM Bachelors FYP]
4. Course coordinator for the following courses
 - ✓ Advanced Nano electronics and Spintronics,
 - ✓ Introduction to MEMS, Analog integrated Circuits,
 - ✓ Mechatronics, Digital Electronics, Bioinformatics,
 - ✓ Electronic Measurements, and Instrumentation
5. Labs Managed
 - ✓ Digital electronics lab
 - ✓ Analog integrated Circuits lab,
 - ✓ Basic electronics lab,
 - ✓ Electronic Design Automation Lab
6. Training Co-coordinator for B.Tech students.
7. Coordinated Workshop on Outcome based Accreditation
8. Assisted in organizing Faculty Effectiveness and Ethical Training (November 2015).
9. Coordinated National Technology Day celebrations, May 2016

ADDITIONAL EXPERIENCE

6. Pioneered IIT-NYCU Dual/Joint Degree program admission which helped in developing comprehensive understanding of Admission procedures and inter university collaborations
7. Installation and maintenance of microfabrication tools like Sputtering and thermal evaporation system at Center for Applied Research in Electronics (CARE), IIT Delhi (IITD).
8. Involved in maintenance & training of fabrication and characterization facilities at central clean room facility, IITD.
9. Designed and built setup for the Pressure measurements for Spintronic pressure sensor.
10. Member of Internal Quality Assurance Cell, College of Engineering Munnar and active involvement in Institutional Development Plan for College of Engineering Munnar (C.E.M.)
11. Represented College of Engineering Munnar at the Workshop on “Strategic Planning for Engineering Colleges” and developed TEQIP Strategic Plan for C.E M.
12. Class tutor and mentor for ECE 2014-2018 batch.
13. Admission Committee member for 2015 and 2016 admissions at College of engineering Munnar

TRAINING PROGRAMS ATTENDED

1. Training on “Advanced Processing Equipments” by **National Nano Device Laboratories**, NAR Labs , Taiwan Semiconductor Research Institute (TSRI) , 10th and 11th February 2020.
2. Workshop on “Strategic Planning for Engineering Colleges” organized by at College of Engineering Vadakara on 18th February 2016
3. Workshop on “National Programme on Technology Enhanced Learning (NPTEL)” organized by IIT Madras at Amal Jyothi College of Engineering, Kanjirappally on 21st November 2016.

REFERENCES

1. **Prof. Yuan-Chieh Tseng**
Distinguished Professor,
Dept. Materials Science &
Engineering, NYCU, 1001 Ta
Hsueh Road, Hsin-Chu-30010,
Taiwan
TEL: (O) +886-35731898
E-mail: yctseng1978@nycu.edu.tw

2. **Prof. Pushparaj Singh**
Associate Professor,
Centre for Applied Research in
Electronics, Indian Institute of
Technology-Delhi, Hauz Khas,
New Delhi-110016, INDIA
TEL: (O) +91-11-26591130
E-mail: prsingh@care.iitd.ac.in

3. **Prof. Lai Changquan**
Assistant Professor,
School of Mechanical &
Aerospace Engineering,
NTU, Singapore -639798,
Singapore
TEL: (O) +65-6790 4952
E-mail: cqlai@ntu.edu.sg